Parsa Oryani

M.Sc. Computer Engineering – Secure Computing, Sharif University of Technology
Research focus: Security of LLM agents and multi-agent systems
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Research Profile

M.Sc. student in Secure Computing at Sharif University of Technology. Current research focuses on the security of LLM agents and multi-agent systems, particularly information-flow control, authorization, provenance, secure memory/tool interactions, and runtime enforcement. Background in cryptography, formal methods, distributed systems, blockchain security, and practical system building.

Education

M.Sc. in Computer Engineering — Secure Computing

Sharif University of Technology, Tehran, Iran
Sep 2025 - Present
National Master's Entrance Examination Rank: **56**

- Research interests: LLM-agent security, information-flow control, authorization, provenance, and secure agent architectures.
- Selected coursework: Applied Cryptography (18.9/20), Deep Learning (18.35/20, Dr. Soleymani), Secure Software Systems (18.2/20), Formal Methods in Information Security, and Foundations and Applications of Blockchain.
- Deep Learning: Transformers, LLMs, state-space and generative models, self-supervised learning, multimodal foundation models, and model evaluation.

B.Sc. in Computer Science

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
Sep 2020 - Feb 2025
National University Entrance Examination Rank: **343**

- Background in cryptography, algorithms, probability, linear algebra, AI, and software development.
- Cryptography I (18.75/20); Special Topics in Cryptography (18.25/20, highest grade in class).

Research Experience

Research Assistant — Agentic AI Security

Sharif University of Technology; Supervisor: Dr. Morteza Amini
Sep 2025 - Present

- Investigating the security of LLM-based agentic and multi-agent systems, including information-flow control, authorization, provenance, memory/state, tool/skill security, and runtime enforcement.
- Studying trust boundaries, least privilege, secure composition, update safety, and recovery from unsafe effects through systems security, program analysis, formal methods, and privacy.
- Previous research covered blockchain security, cross-chain interoperability, Layer-2 systems, state verification, and protocol safety/liveness.

Teaching Experience

Teaching Assistant, Computer Networks

Sharif University of Technology; Dr. Sadeghzadeh — Feb – Jun 2026

Teaching Assistant, Machine Learning

Sharif University of Technology; Dr. Motahari — Feb – Jun 2026

Teaching Assistant, Artificial Intelligence and Lab

Amirkabir University of Technology; Dr. Ghatee — Feb – Jun 2025

Teaching Assistant, Design and Analysis of Algorithms

Amirkabir University of Technology; Dr. Maryam Saneifard — Oct 2023 – Jan 2024

Teaching Assistant, Foundations of Matrices and Linear Algebra

Amirkabir University of Technology; Dr. Najafi — Sep 2023 – Jan 2024

Teaching Assistant, Advanced Programming

Amirkabir University of Technology; Dr. Bejani — Feb – Jun 2023

Selected Projects

ZK-Mixer: Regulated Anonymous Payments — 2026

Research prototype exploring privacy with tiered disclosure for authorized audit scenarios. Case study · Repository

Ethereum CLI (Sepolia Testnet) — 2025

Prototype CLI with encrypted local key storage, scoped network access, and failure-aware RPC handling. Case study · Repository

BRCC: Besu Research Control Center — 2025

Research control plane for safety-gated, reproducible experiments on a local Besu QBFT network. Case study · Repository

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Technical Research Toolkit

Agentic AI & LLM Systems

LLM agents, multi-agent systems, tool/MCP/skill integration, RAG, prompting and inference, agent memory/state, and agent evaluation.

Security & Formal Methods

Threat modeling, information-flow control, authorization and least privilege, provenance, taint/data-flow analysis, security invariants, runtime enforcement, SMT/model checking, and formal verification concepts.

Deep Learning

PyTorch, Transformers and attention, LLMs, state-space models, generative models, self-supervised learning, multimodal models, and model evaluation.

Cryptography & Distributed Systems

Applied cryptography, commitments, Merkle structures, zero-knowledge proof concepts, consensus, blockchain protocols, Layer-2 systems, and cross-chain architectures.

Engineering

Python, Rust, Solidity, C++, PostgreSQL, Linux, Git.

Last updated: September 2026